ASSIGNMENT – 3

Introduction to Symmetric Cryptography

Cryptography and Network Security

Instructions

- Attempt all 8 questions. Marks are indicated against each question.
- Support theoretical answers with relevant examples, algorithm steps and diagrams.
- Use neat, labelled diagrams wherever required.
- Write answers in your own words. Cite any reference books or websites you use.

Questions

Q1. *[Principles and techniques]* **[10 Marks]**

Explain the basic principles of symmetric cryptography. Describe the roles of the plaintext, ciphertext, secret key, encryption algorithm and decryption algorithm, and explain Kerckhoffs's principle. Illustrate your answer with a neat labelled block diagram of a symmetric cryptosystem.

Ans-Introduction

Symmetric cryptography is a method of protecting information using **one secret key**. The same key is used to encrypt and decrypt the data.

Main Components

- **Plaintext:** The original readable message or data.
- **Secret Key:** A private key known only to the sender and receiver.
- **Encryption Algorithm:** Converts plaintext into unreadable ciphertext using the secret key.
- **Ciphertext:** The encrypted and unreadable form of the original message.
- **Decryption Algorithm:** Converts ciphertext back into plaintext using the secret key.

Working

The basic process is:

Plaintext → Encryption + Secret Key → Ciphertext → Decryption + Secret Key → Plaintext

Kerckhoffs's Principle

Kerckhoffs's principle states that the security of a cryptographic system should depend on the **secret key**, not on keeping the encryption algorithm secret. The algorithm can be publicly known, but the key must remain secret.

Q2. *[Principles and techniques]* **[10 Marks]**

Distinguish between stream ciphers and block ciphers. Discuss the concepts of confusion and diffusion, and explain the main block cipher modes of operation (ECB, CBC, CFB, OFB and CTR) along with the strengths and weaknesses of each mode.

Ans- Stream Cipher

A stream cipher encrypts data **one bit or one byte at a time**.

- Fast and suitable for continuous data.
- Requires careful management of the keystream.
- Reusing the same keystream can create serious security problems.

Block Cipher

A block cipher divides data into **fixed-size blocks** and encrypts each block.

Confusion and Diffusion

- **Confusion:** Hides the relationship between the secret key and ciphertext.
- **Diffusion:** Spreads the effect of one plaintext bit over many ciphertext bits.

Block Cipher Modes

- **ECB:** Each block is encrypted separately.
  - Simple and fast.
  - Repeated plaintext blocks produce repeated ciphertext blocks, so it is less secure.
- **CBC:** Each plaintext block is combined with the previous ciphertext block.
  - Hides repeated patterns better.
  - Requires an initialization vector.
- **CFB:** Uses previous ciphertext to create a stream-like encryption.
  - Useful for data arriving in small units.
  - Errors can affect following data.
- **OFB:** Generates a keystream independently of the plaintext.
  - Transmission errors do not spread.
  - Reusing the same keystream is unsafe.
- **CTR:** Uses a counter to generate encryption blocks.
  - Fast and can be processed in parallel.
  - Counter values must never be reused with the same key.

Q3. *[Historical development]* **[10 Marks]**

Trace the historical development of symmetric encryption algorithms. Your answer should cover early ciphers such as the Caesar cipher and the Enigma machine, Shannon's work on information theory, the Feistel cipher, the adoption of DES in 1977, the rise of Triple DES and other algorithms (such as IDEA, Blowfish and RC4), and the open competition that led to AES.

Ans- Early Ciphers

- **Caesar Cipher:** One of the earliest encryption methods. It shifts letters by a fixed number of positions.

- **Enigma Machine:** Used rotating components to create changing substitutions. It was widely used during World War II.

Shannon's Contribution

Claude Shannon's work on information theory provided important concepts for cryptography, especially:

- **Confusion**
- **Diffusion**

These ideas became important in designing strong encryption algorithms.

Feistel Cipher

The **Feistel structure** became an important design for block ciphers. It divides data into parts and processes them through several rounds.

DES

The **Data Encryption Standard (DES)** was adopted in **1977**. It became a widely used symmetric encryption standard.

Other Algorithms

As DES became weaker, alternatives and improvements appeared:

- **Triple DES**
- **IDEA**
- **Blowfish**
- **RC4**

AES

Because stronger security was required, an open competition was held to find a replacement for DES. **Rijndael** was selected and became the **Advanced Encryption Standard (AES)**.

Q4. *[Classical symmetric encryption]* **[10 Marks]**

Explain the following classical substitution ciphers with their working principle, strengths and weaknesses: (a) Caesar cipher, (b) monoalphabetic substitution cipher, (c) Playfair cipher, and (d) Vigenere polyalphabetic cipher. Why is frequency analysis effective against some of these ciphers but not others?

Ans- A. Caesar Cipher

The Caesar cipher replaces each letter by shifting it a fixed number of positions.

Example:
If the shift is 3:

A → D, B → E, C → F

Strengths

- Very simple.
- Easy to understand and implement.

Weaknesses

- Very small number of possible keys.
- Can easily be broken by trying all shifts.

B. Monoalphabetic Substitution Cipher

In this cipher, every plaintext letter is replaced by another fixed letter.

Strengths

- More possible keys than Caesar cipher.
- Simple to implement.

Weakness

- The same plaintext letter always produces the same ciphertext letter.
- Letter patterns remain visible.

C. Playfair Cipher

Playfair encrypts **two letters at a time** using a 5×5 letter matrix.

Strengths

- Encrypts pairs instead of individual letters.
- Hides simple single-letter frequency patterns.

Weakness

- More complicated than Caesar cipher.
- Can still be attacked using language patterns.

D. Vigenere Cipher

The Vigenere cipher uses a **keyword** to apply different Caesar shifts to different letters.

Strengths

- Stronger than simple substitution.
- Makes normal frequency analysis more difficult.

Weakness

- Repeated keywords can create patterns.
- Long-term or repeated-key use can be attacked.

Frequency Analysis

Frequency analysis works well against Caesar and monoalphabetic ciphers because the same substitution is used repeatedly. It is less effective against Vigenere because the substitution changes according to the keyword.

Q5. *[Classical symmetric encryption]* **[8 Marks]**

Describe the classical transposition techniques, including the rail fence cipher and the columnar transposition cipher. Explain how product ciphers combine substitution and transposition, and why this idea forms the foundation of modern block ciphers. Also discuss the one-time pad and explain why it is theoretically unbreakable yet impractical.

Ans- ## Transposition Cipher

A transposition cipher does not change the letters. Instead, it **changes their positions**.

Rail Fence Cipher

- The message is written in a zigzag pattern.
- The letters are then read row by row.
- It is simple but provides weak security.

Columnar Transposition

- The plaintext is written in rows under a keyword.
- Columns are arranged according to the keyword.
- The columns are then read in a specific order.

It is stronger than a simple rail fence cipher but can still be attacked using patterns.

Product Cipher

A product cipher combines:

- **Substitution** – changes the symbols.
- **Transposition** – changes their positions.

Combining these techniques provides stronger confusion and diffusion. This basic idea became important in the design of modern block ciphers.

One-Time Pad

A one-time pad uses a **truly random key as long as the message**.

Advantages

- Theoretically unbreakable when used correctly.
- Provides perfect secrecy.

Disadvantages

- Very long keys are required.
- Keys must be securely shared.
- Keys must never be reused.
- Storing and managing keys is difficult.

Q6. *[Data Encryption Standard (DES)]* **[12 Marks]**

Describe the structure of the Data Encryption Standard (DES). Explain the Feistel structure, block size and key size, the initial and final permutations, the 16 rounds, the role of the round function (expansion, key mixing, S-boxes and P-box permutation), and the key schedule. Support your answer with a neat diagram of one DES round.

Ans- Introduction

DES is a **symmetric block cipher** that encrypts data in blocks of **64 bits**. Its effective key size is **56 bits**.

Feistel Structure

DES uses a **Feistel structure**. The 64-bit plaintext is divided into two 32-bit halves:

- Left half
- Right half

DES performs **16 rounds** of processing.

Initial and Final Permutation

- First, DES applies an **Initial Permutation (IP)** to the plaintext.
- After 16 rounds, a **Final Permutation** is applied.

DES Round Function

The right half goes through several steps:

1. **Expansion:** 32 bits are expanded to 48 bits.
2. **Key Mixing:** The expanded data is XORed with a 48-bit round key.
3. **S-Boxes:** The data is passed through eight S-boxes.
4. **P-Box:** The resulting bits are rearranged.

The output is then combined with the left half.

Key Schedule

The original key is processed to generate **16 different round keys**, one for each round.

Simple DES Round

Right Half → Expansion → XOR with Round Key → S-Boxes → P-Box → XOR with Left Half → New Halves

Q7. *[Data Encryption Standard (DES)]* **[10 Marks]**

Discuss the strength and weaknesses of DES. Cover the avalanche effect, the issue of its short 56-bit key, brute-force attacks, and differential and linear cryptanalysis. Explain how Double DES and Triple DES (2-key and 3-key) were developed to overcome these weaknesses, and why Double DES is vulnerable to the meet-in-the-middle attack.

Ans- Strengths of DES

- DES has a well-studied design.
- It can be implemented efficiently.
- It has a strong **avalanche effect**.

Avalanche Effect

The avalanche effect means that a small change in the plaintext or key can cause many bits of the ciphertext to change. This makes it difficult to predict the output.

Weaknesses of DES

The main weakness is its **56-bit key**.

- Modern computers can perform large numbers of key guesses.
- This makes **brute-force attacks** practical.
- DES has also been studied against **differential cryptanalysis** and **linear cryptanalysis**.

Double DES

Double DES applies DES twice using two keys.

Plaintext → DES → DES → Ciphertext

However, it does not provide the expected security because of the **meet-in-the-middle attack**.

Triple DES

Triple DES applies DES three times.

It can use:

- **2 keys**
- **3 keys**

Triple DES provides much better security than single DES but is slower than modern algorithms such as AES.

Q8. *[Advanced Encryption Standard (AES)]* **[10 Marks]**

Explain the Advanced Encryption Standard (AES). Describe its block size, supported key lengths (128, 192 and 256 bits) and the corresponding number of rounds. Explain the four round transformations (SubBytes, ShiftRows, MixColumns and AddRoundKey), the key expansion process, and how decryption differs from encryption. Compare AES with DES in terms of structure, security and performance.

Ans- Introduction

AES is a modern symmetric block cipher developed as a replacement for DES.

Basic Specifications

- **Block size:** 128 bits
- **Key sizes:** 128, 192 and 256 bits
- **128-bit key:** 10 rounds
- **192-bit key:** 12 rounds
- **256-bit key:** 14 rounds

Four Main Transformations

1. SubBytes

Each byte is replaced using an AES substitution table called an **S-box**.

2. ShiftRows

Bytes in different rows are shifted by different amounts.

3. MixColumns

Data in each column is mathematically mixed to provide diffusion.

4. AddRoundKey

The data is combined with the round key using **XOR**.

The final AES round does not use MixColumns.

Key Expansion

AES expands the original key to create a separate round key for each round.

Decryption

Decryption uses the inverse operations in the reverse order, including:

- InvSubBytes

- InvShiftRows
- InvMixColumns
- AddRoundKey

AES vs DES

- AES supports much larger keys than DES.
- AES provides stronger security.
- AES is faster and more efficient on modern systems.
- DES uses a Feistel structure, while AES uses a substitution-permutation based structure.
- AES is widely preferred for modern secure communication.